Caleb Bettcher | calebbettcher@gmail.com | calebbettcher.com

EDUCATION

Cornell University - Master of Computer Science New York, NY | **Aug 2026 – May 2027**

- Recipient of a Cornell Tech merit scholarship

University of Colorado Boulder - B.S. Computer Science Boulder, CO | **Aug 2023 – May 2026**

- 3.92 Cumulative GPA, Dean's List, Summa Cum Laude, Accelerated 3-year track

Minor in Business – CU Quants Trading Team – CU Triathlon Team & CU Freeride - Member

WORK EXPERIENCE

Chicago Trading Company – Associate Software Engineer Chicago, IL | **June 2027 – Present**

Belvedere Trading – Quantitative Software Engineering Intern Boulder, CO | **Summer 2026**

- Completed Belvedere Trading University (BTU): an intensive crash course in core options pricing theory, delta hedging, and market making principles, including coursework in advanced C++ optimization and Python market making projects
- Built a real-time C++ market data pipeline ingesting live data from 20+ exchanges via third-party normalization, sustaining 30K+ exchange messages/sec, spiking to well over 100K/sec during market volatility with sub-10μs average app-processing latency
- Designed a fully customizable Signal configuration methodology giving Quants complete control over parameters, per-tradable subscription granularity, multi-threaded dispatch, and Protobuf output, all built for easy extension to future Signals
- Implemented and deployed a proprietary logistic model predicting future trade direction using live market data

Handshake AI Solutions – Prompt Engineer / AI Research Fellow **Aug 2025 – June 2026**

- Designed ~20 adversarial prompts accepted into frontier model evaluation datasets for leading AI labs, targeting reasoning gaps in formal verification (SAT/SMT solving) and graph theory, including directed graph reasoning and DPLL/CDCL algorithm application
- Identified and documented over 10 recurring model failure modes including test set leakage, result hallucination, eval script manipulation, and action-ideation mismatch, to improve reliability and interpretability of frontier LLM models
- Promoted to peer reviewer based on output quality and contribution history, evaluating annotators' model failure classifications for rubric adherence and justification accuracy while incorporating direct feedback from AI research leads

Machine Learning Research – University of Colorado Boulder **Aug 2024 – May 2025**

- Developed custom Julia and Python libraries for processing multi-dimensional satellite data spanning over 6 years
- Iteratively refined a custom YOLOv8x model from a 1-class to 4-class training set, leveraging university supercomputing resources to reduce false-positive detections by ~80% while maintaining a 73.6% training set recall
- Presented on the application and optimization of ML/AI to astrophysics at a national research conference

YouTube – Content Creator **Apr 2020 – June 2026**

- Built a YouTube channel with over 80,000 subscribers and 10 Million views across 150 videos, hired an editor to streamline video production, and managed 5-figure annual sponsorship revenue through partnerships with gaming companies such as Supercell

Net-Results Marketing Automation – Full Stack SWE Intern Denver, CO | **Summer 2022 & 2023**

- Constructed React frontend and Laravel/GraphQL backend systems to handle multi-layered JSON mutations, execute user-customized marketing actions, and synchronize data with Salesforce and Dynamics 365 for 20,000+ active users

RESEARCH PROJECTS

Quantitative Trader – CU Quants University of Colorado Boulder | **Jan 2026 – May 2026**

- Monitored live market-making systems across 12 stablecoin instruments on Kraken US and OKX US, maintaining 98.44% uptime across a 1-178 bps spread capture range while making real-time decisions helping achieve the team's 137.63% annualized returns

SurfaceEdge - Novel CNN Options Pricing Model from Surface Images **Jan 2026 – May 2026**

- Engineered a novel multimodal deep learning pipeline combining CNN-encoded options surface images with contract-level scalars across 200M+ labeled contracts from 17 years of historical data, extending prior computer vision research from CU ML lab work
- Designed and compared three model architectures achieving 7.9% improvement in MAE over the naive baseline, while enforcing a strict chronological train/test split to prevent lookahead bias
- Adapted a causal self-attention transformer to model price history as a sequence prediction problem, finishing as our top model

SKILLS & EXPERTISE

- **Languages –** Python, C++, Java, JavaScript, SQL, TypeScript, Julia
- **Frameworks –** PyTorch, YOLO, React, Next.js, Laravel, GraphQL, NumPy, Pandas
- **Technical Skills –** Git, AWS, HPC Clusters, JVM/JIT Optimization, Machine Learning, Computer Vision, Data Analysis